

Hand Gesture Recognition

30 Days Complete Roadmap (Python + MediaPipe)

Tools & Requirements

Python 3.7+ | OpenCV | MediaPipe | NumPy | Webcam

Install: `pip install opencv-python mediapipe numpy`

Week 1: Environment Setup & MediaPipe Basics

Day 1:

- Install Python, OpenCV, MediaPipe, NumPy
- Test webcam access with OpenCV
- Study `hand_gesture.py` file structure

Day 2:

- Learn OpenCV basics: capture video, show frames
- Understand BGR color format
- Draw basic shapes on video frames

Day 3:

- Learn MediaPipe `HandLandmarker` basics
- Understand 21 hand landmarks
- Detect hand landmarks from webcam

Day 4:

- Study landmark IDs: `THUMB_TIP`, `INDEX_TIP`, etc.
- Understand `TIP_IDS`, `DIP_IDS`, `PIP_IDS`, `MCP_IDS`
- Practice identifying each finger point

Day 5:

- Learn `dist()` and `dist_2d()` functions
- Learn `angle_3pts()` function
- Practice measuring distances between landmarks

Day 6:

- Study `is_finger_up()` function
- Study `is_finger_curled()` function
- Practice detecting finger states

Day 7:

- Review Week 1 concepts
- Test hand landmark detection stability
- Fix any setup issues

Week 2: Gesture Detection & Classification

Day 8:

- Study `get_finger_states()` function
- Understand handedness (Left/Right)
- Practice detecting all 5 finger states

Day 9:

- Study `thumb_beside_fist()` function

- Study `thumb_is_across_palm()` function
- Practice detecting A, B, S signs

Day 10:

- Study `fingertips_touching_thumb()` function
- Study `thumb_index_circle()` function
- Practice detecting O, F, D signs

Day 11:

- Study `hand_pointing_sideways()` function
- Study `hand_pointing_down()` function
- Practice detecting G, H, P, Q signs

Day 12:

- Study `index_middle_crossed()` function
- Practice detecting R, U, V, K signs
- Understand `idx_mid_tip` distance for U vs V

Day 13:

- Study `classify_asl_letter()` full function
- Understand the 11 classification groups
- Test each ASL letter A-Z

Day 14:

- Test all 26 ASL letters
- Note which letters are confused
- Document accuracy issues

Week 3: UI & Visualization

Day 15:

- Study `draw_hand_landmarks()` function
- Practice drawing landmarks on video
- Customize landmark colors

Day 16:

- Study `draw_status_panel()` function
- Understand responsive UI scaling
- Practice modifying panel layout

Day 17:

- Study `draw_reference_panel()` function
- Understand `ASL_DESCRIPTIONS` dictionary
- Add custom descriptions

Day 18:

- Study `LetterSmoother` class
- Understand `buffer_size` parameter
- Test smoothing with different buffer sizes

Day 19:

- Study the `main()` function flow
- Understand `HandLandmarkerOptions`
- Practice modifying detection confidence

Day 20:

- Add new UI features (score, timer)
- Customize color scheme
- Add your name to the UI

Day 21:

- Test full UI on different screen sizes
- Fix any UI scaling issues
- Review all visual elements

Week 4: Improvement & Final Project

Day 22:

- Improve classification accuracy for M, N, T
- Fine-tune distance thresholds
- Test edge cases

Day 23:

- Add word building feature
- Collect detected letters into words
- Add backspace gesture

Day 24:

- Add quiz/practice mode
- Show random letter to practice
- Give feedback if correct

Day 25:

- Add score tracking system
- Track correct/incorrect gestures
- Display accuracy percentage

Day 26:

- Performance optimization
- Reduce lag in detection
- Test on different lighting conditions

Day 27:

- Error handling & bug fixes
- Handle webcam not found error
- Handle model file missing error

Day 28:

- Record demo video of the project
- Test all 26 letters on camera
- Document any remaining issues

Day 29:

- Update README.md with new features
- Add screenshots to documentation
- Write project report

Day 30:

- Final review & testing
- Push final version to GitHub
- Submit project!